SouthColours Application — Complete Run & Setup Guide

This document provides a clean, step-by-step developer manual for running the entire SouthColours ecosystem: the **Spring Boot backend**, **MySQL database**, **Customer mobile app**, and **Admin mobile app**.

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1. System Architecture Overview

Component	Technology	Default Port	Description
Backend API	Spring Boot 3.3 / Java 21	`8080`	REST API, JWT Authentication, Hibernate JPA catalog and order management
Database	MySQL 8.0	`3306`	Schema: `southcolours_db`, User: `southcolours_user`
Customer App	React Native / Expo	`8081`	E-commerce shopping experience for mobile users
Admin App	React Native / Expo	`8082`	Store administration portal (product, category, stock management)

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2. Prerequisites & Installed Paths

Make sure the following prerequisites are installed on your machine:

- **Java JDK 21**: Installed at `C:\Program Files\Eclipse Adoptium\jdk-21.0.12.101-hotspot`
- **Apache Maven 3.9**: Installed at `C:\Users\jadet\maven\apache-maven-3.9.6\bin\mvn.cmd`
- **Node.js (v20+ / v24)**: Installed system-wide
- **MySQL Server 8.0**: Installed as a Windows Service running on port `3306`
- **Expo Go App**: Installed on your physical Android / iOS device (from Google Play Store / Apple App Store)

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3. Step-by-Step Running Guide

Step 1: Start the MySQL Database Service

Ensure the local MySQL service is active:

```
Get-Service -Name MySQL80 | Start-Service
```

Database Credentials:

- Database Name: `southcolours_db`
- Username: `southcolours_user`
- Password: `southcolours_password`

Step 2: Start the Spring Boot Backend Server

Open a terminal in the project root:

1. **Check & Free Port 8080 (if previously occupied):**

```
Get-Process -Id (Get-NetTCPConnection -LocalPort 8080
```

2. **Run Spring Boot:**

```
cd backend
mvn spring-boot:run
```

3. **Verify Backend Health:** Open your browser or run:

```
Invoke-RestMethod -Uri 'http://localhost:8080/api/v1/
```

You should receive a JSON response with `"success": true` and the 8 live categories.

Step 3: Run the Customer Mobile App

Open a **new** terminal tab:

```
cd apps/customer-mobile
npx expo start --clear
```

- **Running on Physical Device (Recommended):**
  - Make sure your mobile phone is connected to the **same Wi-Fi network** as your computer.
  - Open the **Expo Go** app on your phone.
  - Scan the QR code displayed in the terminal.
- **Running on Android Emulator:**
  - Press `a` in the terminal once the Metro bundler is ready.

Step 4: Run the Admin Mobile App

Open another **new** terminal tab:

```
cd apps/admin-mobile
```

```
npx expo start --port 8082 --clear
```

- Scan the QR code with **Expo Go** to access the Admin dashboard.
- From here, you can add products, customize categories, edit stock variants, and manage orders.

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4. Connecting Physical Devices over Wi-Fi (LAN)

Both mobile applications dynamically detect your computer's local Wi-Fi IP (e.g. `192.168.1.63:8080`).

If your phone displays a network timeout or cannot reach the backend:

1. **Verify your computer's current Wi-Fi IP**:

   ```
   ipconfig
   ```

2. **Allow inbound traffic on port 8080 through Windows Defender Firewall**:

   ```
   New-NetFirewallRule -DisplayName "Spring Boot 8080" -
   ```

3. Make sure your PC network profile is set to **Private** rather than Public so local LAN devices can communicate.

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5. Troubleshooting & FAQ

Problem	Cause	Quick Fix

`Port 8080 was already in use`	A prior Java process is still holding port 8080 in the background.	Run: `Get-Process -Id (Get-NetTCPConnection -LocalPort 8080).OwningProcess | Stop-Process -Force`
`Network Error` / `Failed to fetch` on phone	Phone is not on the same Wi-Fi, or firewall blocked port 8080.	Check Wi-Fi connection, verify IP address, and add Windows Firewall rule for port 8080.
Changes not updating in Expo app	Expo Metro bundler cache is stale.	Stop Metro (`Ctrl + C`) and restart with cache reset: `npx expo start -c`
Metro port conflict (Port 8081 busy)	Running customer and admin app on the same default port.	Specify a distinct port for admin app: `npx expo start --port 8082`

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6. How to Save as a PDF (Whenever Needed)

If you or your team ever want this guide as a formatted PDF:

1. Open this file in your editor or browser (e.g., in VS Code / IDE with Markdown Preview).
2. Right-click → **Print** (or `Ctrl + P`).
3. Select **"Save as PDF"** / **"Microsoft Print to PDF"**.